

Bi-weekly Report

Report Number: 02

Group No: 14

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Project title Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

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Objectives for this report

1. Property analysis and identification of the operating conditions.
2. Process synthesis and development of the process flow diagram.
3. Flowsheet development in Aspen Plus.
4. Material and energy balance using Aspen Plus.

1 Property Analysis and Operating Conditions

1.1 Property model

The property method is **NRTL-RK**, selected and validated in Report 1 Section 6. HYDRO-01 and NITRO-02 are declared as Henry components under the list HC-1, which is attached to every block, since both are supercritical at the flash and decanter conditions. The ANILINE–WATER pair uses our own three-parameter regression R-DR-1 from Report 1 Section 6.4, namely $A_{ij} = -4.62238$, $A_{ji} = 4.92646$, $B_{ij} = 1433.47$, $B_{ji} = 0$ and $C_{ij} = 0.2$. That fit has been made on liquid–liquid data covering 39–70 °C, which brackets the 40 °C at which DEC and FLASH operate, so the decanter is computed inside the range of the data.

Component i	Component j	Source	Temp. Units	AU	AJ	BIJ	BJI	CU	DU	EU	EJ	FU	FJ	TLOWER	TUPPER
ANILINE	NITRO-01	APV140 VLE-RK	C	0	0	-217.928	287.592	0.3	0	0	0	0	0	83	99
ANILINE	WATER	R-DR-1	C	-4.62238	4.92646	1433.47	0	0.2	0	0	0	0	0	99	168
ANILINE	BENZE-01	APV140 VLE-RK	C	1.9429	-2.8485	-717.708	1302.27	0.3	0	0	0	0	0	25	180.5
ANILINE	CYCLO-01	APV140 VLE-RK	C	3.0918	-2.1102	-910.317	617.869	0.3	0	0	0	0	0	70.89	181.45
NITRO-01	WATER	APV140 LLE-AS...	C	-5.1549	5.8547	2270.62	229.497	0.2	0	0	0	0	0	0	106.6
NITRO-01	BENZE-01	APV140 VLE-RK	C	0.3005	1.446	40.2665	-653.816	0.3	0	0	0	0	0	70	211

Figure 1: Binary interaction parameters as loaded in the simulation. The ANILINE–WATER row carries the source R-DR-1, that is our own regression from Report 1; every other pair remains on the Aspen databank.

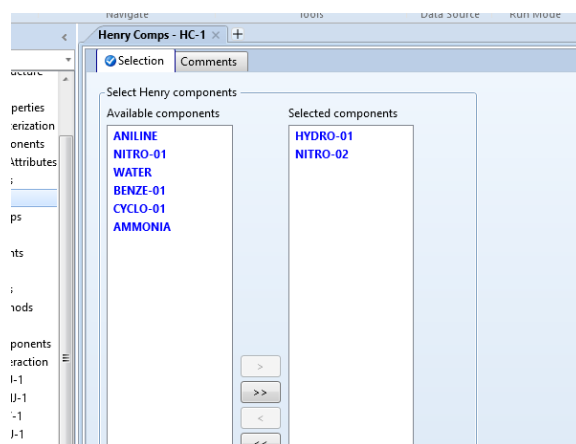


Figure 2: Henry component list HC-1, with HYDRO-01 and NITRO-02 selected.

1.2 Component slate

Aspen ID	Component	Formula	M	T_b (°C)	Role
HYDRO-01	Hydrogen	H_2	2.02	−252.9	Reactant, Henry component, recycled
NITRO-02	Nitrogen	N_2	28.01	−195.8	Inert in the make-up, Henry component
AMMONIA	Ammonia	NH_3	17.03	−33.3	By-product of hydrogenolysis
BENZE-01	Benzene	C_6H_6	78.11	80.1	By-product of hydrogenolysis
WATER	Water	H_2O	18.02	100.0	Main by-product
CYCLO-01	Cyclohexylamine	$C_6H_{13}N$	99.17	134.5	By-product of over-hydrogenation
ANILINE	Aniline	C_6H_7N	93.13	184.1	Product
NITRO-01	Nitrobenzene	$C_6H_5NO_2$	123.11	210.8	Feed, recycled from COL-2 bottoms

The boiling points fix the separation sequence. Aniline and water form a heterogeneous azeotrope at $x_{\text{aniline}} = 0.0368$ and 99.34 °C, so no column can separate the pair and a decanter is used instead. The 26.7 K gap between aniline and nitrobenzene is the split which carries the 5 ppm nitrobenzene limit, and it is therefore placed last, where the feed is dry and the load is smallest.

1.3 Operating conditions

Unit	Condition	Basis
VAP-1	3 bar, VF = 1	The reaction section is held at one pressure so that no compression is required on the organic side. The block delivers saturated vapour at 260.5 °C.
FEHE	Cold outlet 290 °C	A 10 K approach to the reactor temperature, recovering heat from the effluent.
REACT	300 °C, 3 bar	Operating window of the supported copper catalyst. Below about 250 °C the rate is inadequate; above about 350 °C over-hydrogenation accelerates and the catalyst sinters. Low pressure is preferred as the main reaction loses moles.
COOL-1	120 °C	Staged, so that this duty is available at a temperature at which steam can be raised.
COOL-2	40 °C	Cooling-water approach; condenses the organics ahead of the flash.
FLASH	40 °C, 2.5 bar	0.5 bar below the reaction pressure for line and exchanger losses. The low temperature drives the hydrogen recovery.
SPLIT	Purge fraction 0.0086	Fixed from the hydrogen and nitrogen balances of Section 4.3.
COMP	3.2 bar, $\eta_{is} = 0.75$	Restores the recycle gas above the mixer pressure.
DEC	40 °C, 1.5 bar, key WATER	As cold as the cooling water permits, since the miscibility gap widens as the temperature falls.
COL-1	1.0 bar top	Atmospheric. The overhead is the water-rich azeotrope near 99 °C.
COL-2	0.2 bar top	Vacuum. At atmospheric pressure the nitrobenzene bottoms would sit near 211 °C, at which aniline darkens; at 0.3 bar the reboiler runs at 144 °C.
STRIP	1.0 bar	Atmospheric stripping of the aqueous phase, reboiler at 101 °C.

2 Process Synthesis and Process Flow Diagram

2.1 Synthesis decisions

Continuous operation. 100,000 t/y at 8000 h/y is far above the threshold at which batch operation is defensible.

Input–output structure. Nitrobenzene and hydrogen enter; aniline, a water-rich overhead, treated wastewater and a purge gas leave. A purge is required because the make-up hydrogen carries 0.1 mol% nitrogen which does not react.

Recycle structure. Two reactants of very different volatility give two recycles, a gas recycle for the excess hydrogen and a liquid recycle for the unconverted nitrobenzene. Hydrogen is fed in large excess, which suppresses the aniline-consuming side reactions and acts as thermal ballast for the 17.6 MW exotherm. A third recycle, the recovered aniline from the stripper, follows once the separation system is fixed.

Separation system. The vapour–liquid split is taken first and for free in the flash, and the liquid–liquid split for free in the decanter; only the residue is handed to distillation. Within the distillation train the most plentiful component is removed early, water at 282 kmol/h against 133 kmol/h of aniline, and the highest-purity split is left last.

Heat integration. The feed–effluent heat exchanger is placed at this stage because the reactor feed must be raised to 290 °C and the effluent cooled from 300 °C. The remainder of the network is taken up in Report 5.

2.2 Separation sequence

Because the aniline–water azeotrope is heterogeneous, DEC breaks it and therefore comes first, delivering an aniline-rich organic phase and an aqueous phase of water with dissolved aniline. The organic phase is still wet at its saturation limit, so COL-1 takes the water overhead as the azeotrope; it is specified as a three-phase column with WATER as key over stages 2–19, because the overhead condensate itself splits into two liquid phases. The dry bottoms goes to COL-2, where aniline is taken overhead as product and nitrobenzene leaves in the bottoms and returns to the feed. The aqueous phase is stripped in STRIP, the recovered aniline returning to COL-1 and the stripped water leaving as WASTEWTR.

2.3 Process flow diagram

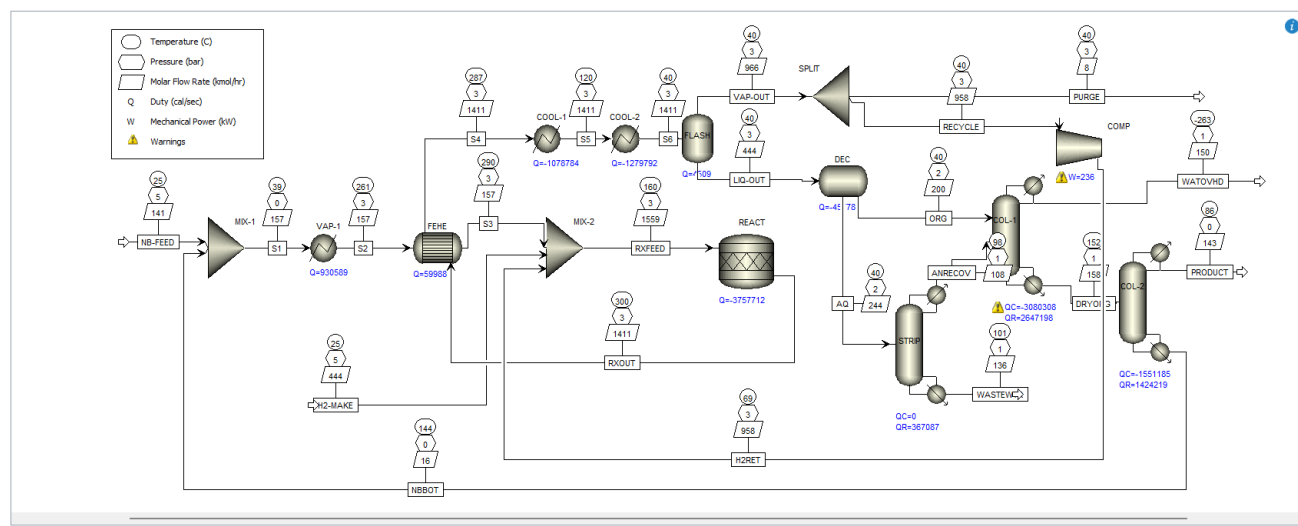


Figure 3: Process flow diagram of the aniline plant as built and converged in Aspen Plus, annotated with temperature (°C), pressure (bar), molar flow (kmol/h), block duty (cal/s) and shaft power (kW).

3 Flowsheet Development in Aspen Plus

3.1 Blocks and specifications

Block	Model	Specification as entered
MIX-1	Mixer	Fresh nitrobenzene with the NBBOT recycle
VAP-1	Heater	3 bar, vapour fraction 1
FEHE	HeatX	Cold outlet 290 °C, minimum approach 1 K
MIX-2	Mixer	Vaporised feed with H2-MAKE and H2RET
REACT	RStoic	300 °C, 3 bar, three reactions, series
COOL-1	Heater	120 °C, $\Delta P = -0.2$ bar
COOL-2	Heater	40 °C, $\Delta P = -0.2$ bar
FLASH	Flash2	40 °C, 2.5 bar
SPLIT	FSplit	Split fraction to PURGE 0.0086
COMP	Compr	Isentropic, 3.2 bar, $\eta_{is} = 0.75$
DEC	Decanter	40 °C, 1.5 bar, key component WATER
COL-1	RadFrac	20 stages, 3 phases, total condenser, kettle reboiler
COL-2	RadFrac	15 stages, 2 phases, total condenser, kettle reboiler
STRIP	RadFrac	8 stages, no condenser, kettle reboiler

3.2 Reaction specification

The three reactions are entered in RStoic with the *series* option enabled, so that each acts on the stream leaving the one before it. Both side reactions consume aniline rather than nitrobenzene, so

unless they follow the main reaction there is no aniline for them to act upon.

No.	Reaction	Conversion, basis	Extent (kmol/h)
1	$\text{C}_6\text{H}_5\text{NO}_2 + 3 \text{H}_2 \rightarrow \text{C}_6\text{H}_5\text{NH}_2 + 2 \text{H}_2\text{O}$	0.99 of NITRO-01	141.113
2	$\text{C}_6\text{H}_5\text{NH}_2 + 3 \text{H}_2 \rightarrow \text{C}_6\text{H}_{13}\text{N}$	0.01513 of ANILINE	2.367
3	$\text{C}_6\text{H}_5\text{NH}_2 + \text{H}_2 \rightarrow \text{C}_6\text{H}_6 + \text{NH}_3$	0.03530 of ANILINE	5.438

The two fractional conversions have been back-calculated so that, applied in series, they reproduce the 95 / 1.5 / 3.5 selectivity split adopted in Report 1.

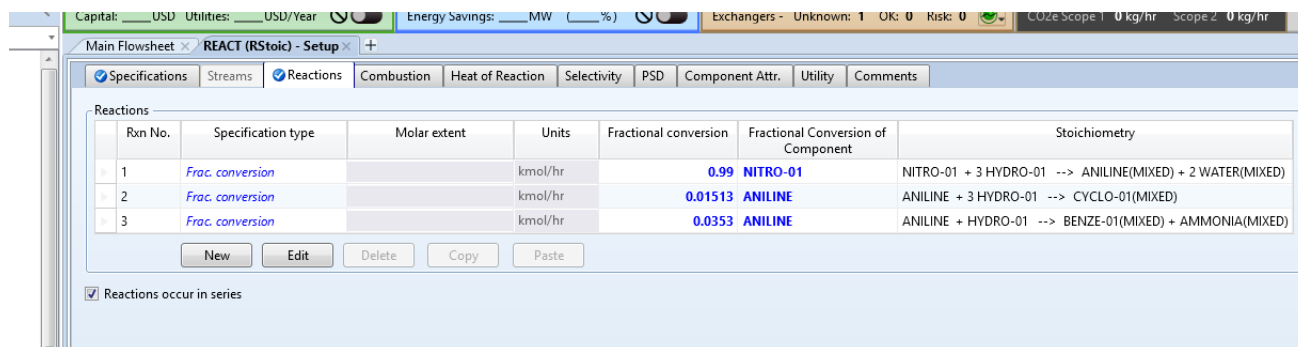


Figure 4: Reaction set entered in the RStoic block, with the three stoichiometries, the fractional conversions and their basis components, and the series option selected.

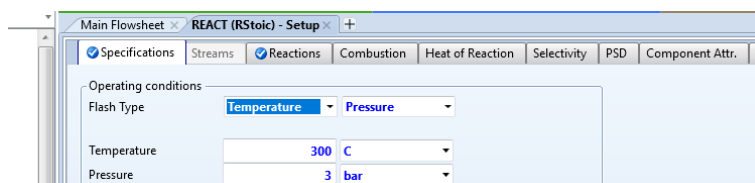


Figure 5: Reactor operating specification, 300 °C and 3 bar.

3.3 Column specifications and results

	COL-1	COL-2	STRIP
Stages	20	15	8
Feed stage	10, above-stage	6, on-stage	1, on-stage
Condenser	Total	Total	None
Reboiler	Kettle	Kettle	Kettle
Convergence	Vapor-Liquid-Liquid	Vapor-Liquid	Standard
Operating specifications	D 150 kmol/h, RR 3	D:F 0.9, RR 2	Boilup ratio 1.0
Top pressure, bar	1.0	0.2	1.0
Reboiler pressure, bar	1.1	0.3	1.05
Distillate, kmol/h	150.00	142.58	108.21
Bottoms, kmol/h	158.42	15.84	136.05
Boilup ratio	5.45	28.67	1.00
Reboiler temperature, °C	151.89	144.15	101.13

3.4 Convergence

Three streams are torn, one for each recycle. The maximum relative errors reported by Aspen Plus are 8.23×10^{-4} on H2RET and 2.66×10^{-5} on S3. The overall mass balance closes to 0.7 kg/h in 18,300 kg/h and the energy balance to 0.003 %, as given in Section 4.

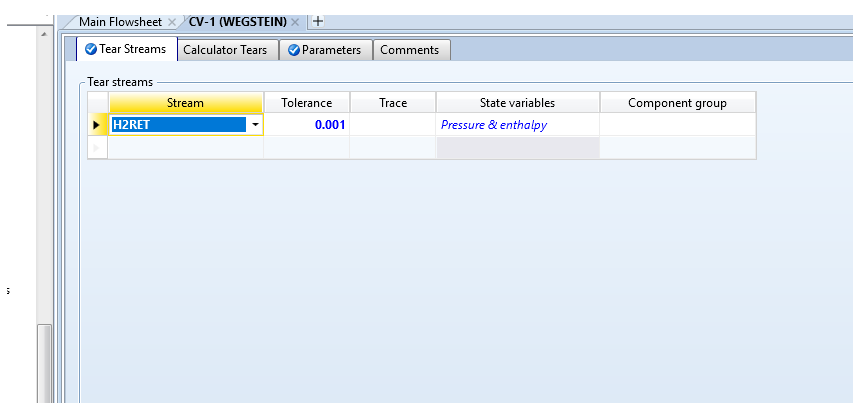


Figure 6: Tear stream definition for the hydrogen loop, converged on pressure and enthalpy to a tolerance of 0.001 by the Wegstein method.

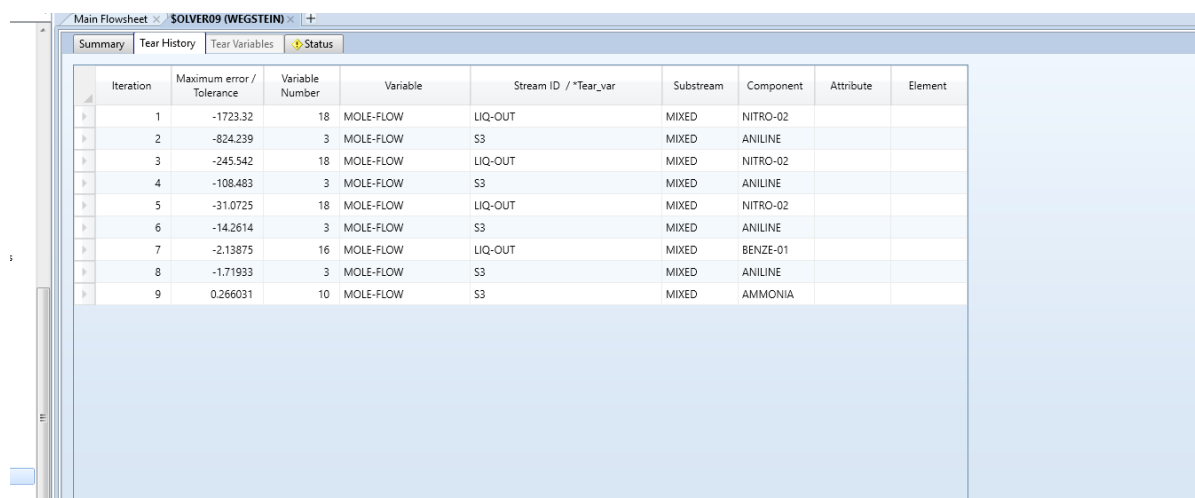


Figure 7: Convergence history, showing the maximum error on each torn variable at the final iteration.

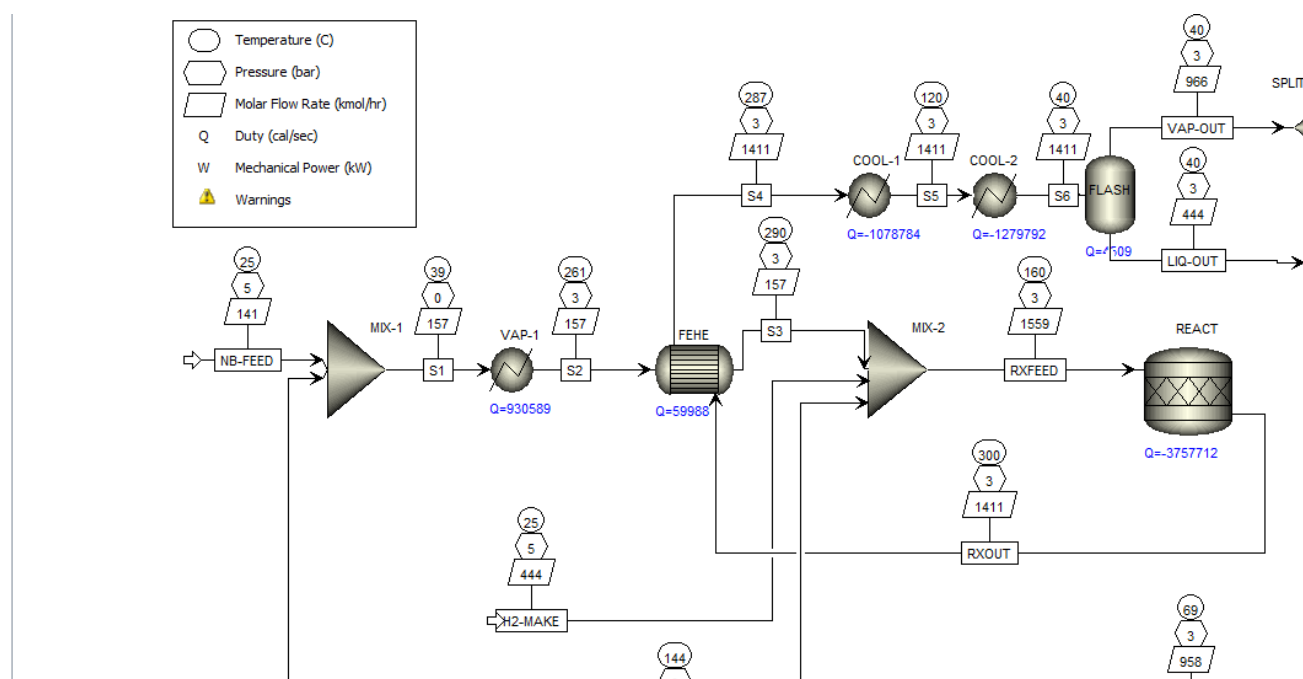


Figure 8: Reaction and hydrogen recycle section of the converged flowsheet.

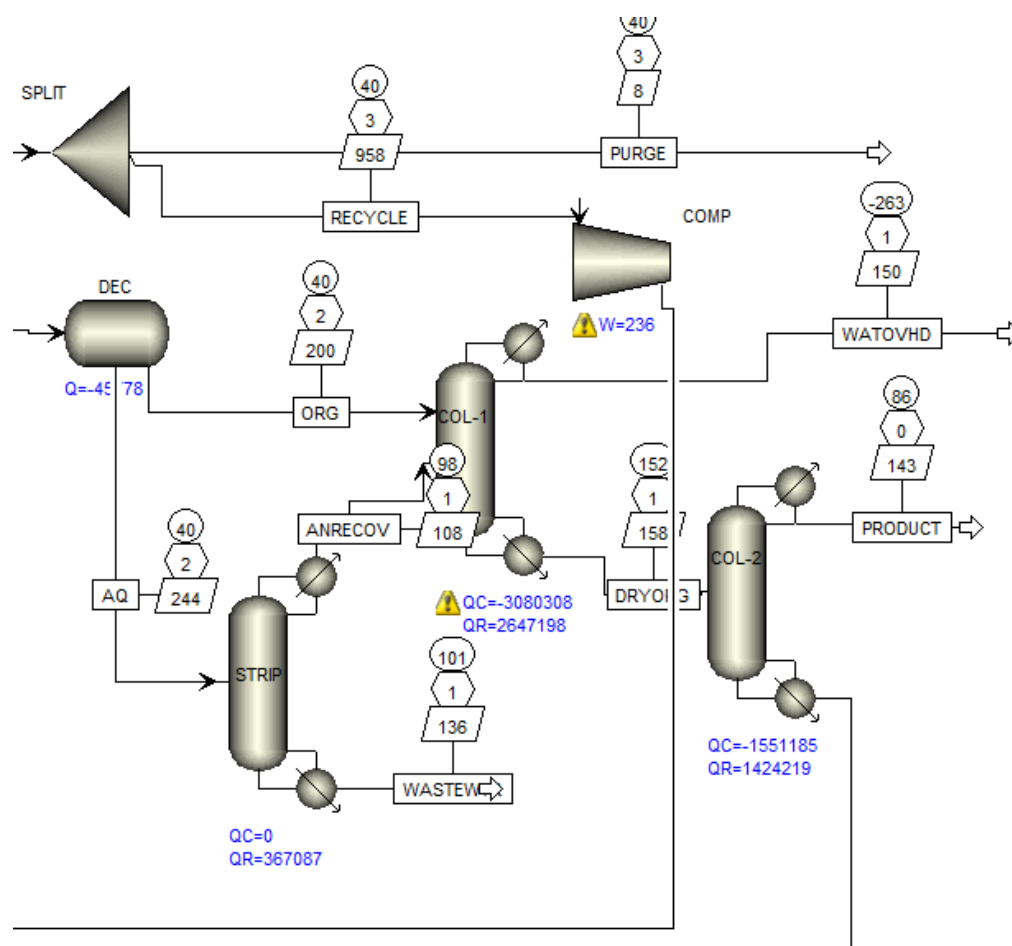


Figure 9: Separation section of the converged flowsheet, with the condenser and reboiler duties of the three columns.

4 Material and Energy Balance

4.1 Overall material balance

In	kmol/h	kg/h	Out	kmol/h	kg/h
NB-FEED	141.28	17,393.2	PRODUCT	142.58	12,346.8
H2-MAKE	444.00	906.6	WATOVHD	150.00	3,372.8
			WASTEWTR	136.05	2,536.8
			PURGE	8.31	42.7
Total		18,299.7	Total		18,299.1

The balance closes to 0.7 kg/h, that is 0.004 %. The converged fresh nitrobenzene feed of 141.28 kmol/h and the hydrogen make-up of 444.0 kmol/h agree exactly with the independent hand balance of Report 1 Sections 3.3 and 3.5.

4.2 Reactor performance

Quantity	Report 1 (hand)	Aspen (converged)
Nitrobenzene to reactor, kmol/h	142.71	142.54
Nitrobenzene reacted, kmol/h	141.28	141.11
Hydrogen at reactor inlet, kmol/h	1427.1	1300.6
Hydrogen-to-nitrobenzene ratio	10.0	9.12
Hydrogen consumed, kmol/h	435.15	435.88
Net aniline produced, kmol/h	134.22	133.31

4.3 Hydrogen loop

Term	kmol/h	Remark
Hydrogen in the make-up	443.56	99.9 mol% in 444.0 kmol/h
Consumed by reaction	435.88	Three reactions
Removed in the purge	7.44	0.86 % split of the flash vapour
Dissolved in the flash liquid	0.13	Computed by Henry's law

The purge fraction follows from the requirement that, at steady state, the hydrogen leaving in the purge equal the make-up less what the reaction consumes and less what dissolves:

$$f_{\text{purge}} = \frac{M - C - D}{V} = \frac{443.56 - 435.88 - 0.13}{864.57} = 0.0087 \quad (1)$$

where V is the hydrogen in the flash vapour. The value used in SPLIT is 0.0086. The recycle gas settles at 89.5 mol% hydrogen with 5.2 mol% nitrogen, which is why the reactor ratio holds at 9.12:1 rather than at the 10:1 assumed in Report 1.

4.4 Stream conditions

Stream	T (°C)	P (bar)	Vapour fraction	kmol/h	kg/h
NB-FEED	25.0	5.00	0.000	141.28	17393.2
H2-MAKE	25.0	5.00	1.000	444.00	906.6
S1	38.8	0.30	0.000	157.12	18906.0
S2	260.5	3.00	1.000	157.12	18906.0
S3	290.0	3.00	1.000	157.12	18906.0
RXFEED	159.7	3.00	0.990	1559.05	24734.9
RXOUT	300.0	3.00	1.000	1410.84	24734.9
S4	287.2	3.00	1.000	1410.84	24734.9
S5	120.0	2.80	0.920	1410.84	24734.9
S6	40.0	2.60	0.684	1410.84	24734.9
VAP-OUT	40.0	2.50	1.000	966.37	4965.9
PURGE	40.0	2.50	1.000	8.31	42.7
RECYCLE	40.0	2.50	1.000	958.06	4923.2
H2RET	69.5	3.20	1.000	957.93	4922.3
LIQ-OUT	40.0	2.50	0.000	444.47	19769.2
ORG	40.0	1.50	0.000	200.21	15088.7
AQ	40.0	1.50	0.000	244.27	4680.5
ANRECOV	98.4	1.00	1.000	108.21	2143.7
WATOVHD	—	1.00	0.000	150.00	3372.8
DRYORG	151.9	1.10	0.000	158.42	13859.6
PRODUCT	86.2	0.20	0.000	142.58	12346.8
NBBOT	144.2	0.30	0.000	15.84	1512.9
WASTEWTR	101.1	1.05	0.000	136.05	2536.8

4.5 Component molar flows — reaction section (kmol/h)

	NB-FEED	H2-MAKE	RXFEED	RXOUT	VAP-OUT	RECYCLE	PURGE	LIQ-OUT
Aniline	—	—	15.316	148.624	0.732	0.726	0.006	147.894
Nitrobenzene	141.280	—	142.538	1.425	0.007	0.007	tr	1.418
Water	—	—	34.396	316.622	34.704	34.406	0.298	281.917
Benzene	—	—	13.660	19.098	13.784	13.665	0.119	5.314
Cyclohexylamine	—	—	0.019	2.385	0.019	0.019	tr	2.367
Nitrogen	—	0.444	50.561	50.561	50.553	50.118	0.435	0.009
Hydrogen	—	443.556	1300.578	864.701	864.568	857.132	7.435	0.134
Ammonia	—	—	1.984	7.423	2.001	1.984	0.017	5.421
Total	141.28	444.00	1559.05	1410.84	966.37	958.06	8.31	444.47

Main Flowsheet × REACT (RStoiC) - Stream Results (Boundary) × +												
Material	Heat	Load	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids					
				Units	NB-FEED ▾	H2-MAKE ▾	RXFEED ▾	RXOUT ▾	VAP-OUT ▾	RECYCLE ▾	PURGE ▾	LIQ-OUT ▾
▶	Mass Density			gm/cc	1.19849	0.000410615	0.00133592	0.00110454	0.000492917	0.000492917	0.000492917	1.06831
▶	Enthalpy Flow			cal/sec	117542	145.752	771308	-2.9864e+06	-454264	-450357	-3906.67	-4.94619e+06
▶	Average MW				123.111	2.04188	15.8654	17.5321	5.13878	5.13878	5.13878	44.4777
▶	— Mole Flows			kmol/hr	141.28	444	1559.05	1410.84	966.368	958.057	8.31077	444.475
▶	ANILINE			kmol/hr	0	0	15.3163	148.624	0.732186	0.725889	0.0062968	147.894
▶	NITRO-01			kmol/hr	141.28	0	142.538	142538	0.00707465	0.0070138	6.0842e-05	1.41831
▶	WATER			kmol/hr	0	0	34.3965	316.622	34.7044	34.406	0.298458	281.917
▶	BENZE-01			kmol/hr	0	0	13.6596	19.098	13.7839	13.6654	0.118542	5.31414
▶	CYCLO-01			kmol/hr	0	0	0.0186126	2.38538	0.0187612	0.0185998	0.000161346	2.36666
▶	NITRO-02			kmol/hr	0	0.444	50.5613	50.5613	50.5527	50.1179	0.434753	0.00868206
▶	HYDRO-01			kmol/hr	0	443.556	1300.58	864.701	864.568	857.132	7.43528	0.133881
▶	AMMONIA			kmol/hr	0	0	1.98445	7.42285	2.00145	1.98423	0.0172124	5.42146
▶	— Mole Fractions											
▶	ANILINE				0	0	0.00982413	0.105344	0.000757668	0.000757668	0.000757668	0.332739
▶	NITRO-01				1	0	0.091426	0.00101031	7.32086e-06	7.32086e-06	7.32086e-06	0.00319097
▶	WATER				0	0	0.0220624	0.224421	0.0359122	0.0359122	0.0359122	0.634271
▶	BENZE-01				0	0	0.00876145	0.0135366	0.0142637	0.0142637	0.0142637	0.011956
▶	CYCLO-01				0	0	1.19384e-05	0.00169075	1.94141e-05	1.94141e-05	1.94141e-05	0.00532462
▶	NITRO-02				0	0.001	0.0324308	0.0358378	0.052312	0.052312	0.052312	1.95333e-05
▶	HYDRO-01				0	0.999	0.83421	0.612898	0.894657	0.894657	0.000301211	
▶	AMMONIA				0	0	0.00127286	0.0052613	0.0020711	0.0020711	0.0020711	0.0121975
▶	+ Mass Flows			kg/hr	17393.2	906.594	24734.9	24734.9	4965.95	4923.24	42.7072	19769.2
▶	+ Mass Fractions											
▶	Volume Flow			l/min	241.875	36798.2	308589	373232	167910	166466	1444.03	308.419

Figure 10: Aspen Plus stream results for the reaction section, as tabulated above.

4.6 Component molar flows — separation section (kmol/h)

	ORG	AQ	ANRECOV	WATOVHD	DRYORG	PRODUCT	NBBOT	WASTEWTR
Aniline	145.366	2.528	2.470	4.587	143.248	128.657	14.591	0.058
Nitrobenzene	1.416	0.002	0.002	0.101	1.317	0.066	1.251	—
Water	45.206	236.712	101.719	134.393	12.532	12.532	—	134.993
Benzene	5.296	0.018	0.018	5.314	—	—	—	—
Cyclohexylamine	1.222	1.145	0.142	0.041	1.323	1.323	tr	1.003
Nitrogen	0.008	0.001	0.001	0.009	—	—	—	—
Hydrogen	0.113	0.021	0.021	0.134	—	—	—	—
Ammonia	1.583	3.839	3.839	5.421	—	—	—	tr
Total	200.21	244.27	108.21	150.00	158.42	142.58	15.84	136.05

Main Flowsheet × REACT (Stoic) - Stream Results (Boundary) × +											
Material	Heat	Load	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids				
		Units	ORG	AQ	ANRECOV	WATOVHD	DRYORG	PRODUCT	NBBOT	WASTEWTR	
▶	Mass Density	gm/cc	1.06849	0.976142	0.00064614	1.25661	0.959244	1.02292	0.972529	0.913594	
▶	Enthalpy Flow	cal/sec	-497384	-4.49458e+06	-1.6117e+06	-2.85966e+06	317490	133746	56780.2	-2.5158e+06	
▶	Average MW		75.3646	19.1617	19.8104	22.4851	87.4863	86.5963	95.496	18.6458	
▶	— Mole Flows	kmol/hr	200.209	24.265	108.211	150	158.421	142.578	15.8421	13.6054	
▶	ANILINE	kmol/hr	145.366	2.52769	2.46954	4.58746	143.248	128.657	14.591	0.0581526	
▶	NITRO-01	kmol/hr	1.41597	0.00233571	0.00233571	0.101132	1.31717	0.0661604	1.25101	5.9517e-13	
▶	WATER	kmol/hr	45.2056	236.712	101.719	134.393	12.532	12.532	3.26198e-15	134.993	
▶	BENZE-01	kmol/hr	5.29575	0.0183881	0.0183881	5.31414	5.09288e-15	0	0	3.2614e-25	
▶	CYCLO-01	kmol/hr	1.22172	1.14494	0.141849	0.040595	1.32297	1.32297	8.99124e-09	1.00309	
▶	NITRO-02	kmol/hr	0.00784118	0.000840872	0.000840872	0.00868206	1.08364e-49	0	0	1.04848e-41	
▶	HYDRO-01	kmol/hr	0.113065	0.020816	0.020816	0.133881	3.56345e-52	0	0	9.84816e-39	
▶	AMMONIA	kmol/hr	1.5828	3.83866	3.83866	5.42146	5.77084e-12	5.77084e-12	4.67493e-37	2.05157e-06	
▶	— Mole Fractions										
▶	ANILINE		0.726073	0.0103481	0.0228214	0.0305831	0.904229	0.902362	0.921032	0.000427423	
▶	NITRO-01		0.00707246	9.56217e-06	2.15847e-05	0.000674216	0.00831441	0.000464028	0.0789679	4.37452e-15	
▶	WATER		0.225792	0.969076	0.940002	0.895951	0.0791056	0.0878951	2.05906e-16	0.9922	
▶	BENZE-01		0.0264511	7.52792e-05	0.000169927	0.0354276	3.21479e-17	0	0	2.39714e-27	
▶	CYCLO-01		0.0061022	0.00468729	0.00131085	0.000270633	0.008351	0.00927889	5.67555e-10	0.00737277	
▶	NITRO-02		3.9165e-05	3.44245e-06	7.77063e-06	5.78804e-05	6.84025e-52	0	0	7.70636e-44	
▶	HYDRO-01		0.000564732	8.5219e-05	0.000192365	0.000892537	2.24936e-54	0	0	7.23842e-41	
▶	AMMONIA		0.00790572	0.0157151	0.0354737	0.0361431	3.64274e-14	4.04748e-14	2.95096e-38	1.50791e-08	
▶	♦ Mass Flows	kg/hr	15088.7	4680.54	2143.71	3372.77	13859.6	12346.8	1512.85	2536.83	
▶	♦ Mass Fractions										
▶	Volume Flow	l/min	235.359	79.9156	55295.4	44.7338	240.808	201.169	25.9264	46.2793	

4.7 Product stream

Component	kmol/h	mol%	wt%
Aniline	128.657	90.24	97.04
Water	12.532	8.79	1.83
Cyclohexylamine	1.323	0.93	1.06
Nitrobenzene	0.066	0.05	0.07
PRODUCT	142.58	100.00	100.00

PRODUCT carries 11,982 kg/h of aniline, that is 11.98 t/h against the design rate of 12.5 t/h. Of the 133.31 kmol/h of aniline produced in the reactor, 128.66 kmol/h reaches the product, giving a recovery of 96.5 % through the separation train.

4.8 Energy balance

Block	Service	cal/s	MW
VAP-1	Vaporiser	+930,589	+3.894
FEHE	Feed-effluent exchanger	59,988	0.251
REACT	Reactor, isothermal at 300 °C	−3,757,712	−15.722
COOL-1	Effluent cooler to 120 °C	−1,078,784	−4.514
COOL-2	Effluent cooler to 40 °C	−1,279,792	−5.354
FLASH	Flash drum	+4,509	+0.019
DEC	Decanter	−45,778	−0.192
COL-1	Condenser	−3,080,308	−12.888
COL-1	Reboiler	+2,647,198	+11.076
COL-2	Condenser	−1,551,185	−6.490
COL-2	Reboiler	+1,424,219	+5.959
STRIP	Reboiler	+367,087	+1.536
COMP	Shaft work	+56,481	+0.236

Taking the FEHE as an internal transfer, the block duties and the compressor work sum to −22.441 MW, against −22.440 MW obtained from the enthalpy flows of the terminal streams, so the energy balance closes to 0.003 %.

The reactor duty of −15.722 MW is the heat of reaction less the sensible heat required to raise the feed from 159.7 to 300 °C:

$$Q_{\text{REACT}} = -(141.113 \times 443 + 2.367 \times 193 + 5.438 \times 51) + Q_{\text{sensible}} = -17.569 + 1.847 = -15.722 \text{ MW} \quad (2)$$

The reaction term of 17.569 MW agrees with the 17.568 MW computed by hand in Report 1 Section 3.6.

5 Results and Discussion

The flowsheet converges with all three recycles closed. The two feed rates reproduce the Report 1 hand balance exactly, the mass balance closes to 0.004 % and the energy balance to 0.003 %, so the simulation and the hand calculation of the previous fortnight are consistent with one another.

The product leaves COL-2 at 97.04 wt% aniline. The residual water at 1.83 wt% and cyclohexylamine at 1.06 wt% are both lighter than aniline and therefore leave overhead with it, so a light-ends draw at the top of COL-2 is the route to the 99.9 wt% specification. Nitrobenzene stands at 660 ppm against a limit of 5 ppm, which calls for a sharper cut in the same column. Both are taken up in the distillation column design of Report 3.

COL-2 returns 14.59 kmol/h of aniline in NBBOT along with 1.25 kmol/h of nitrobenzene, so the reactor feed carries 15.32 kmol/h of aniline before reaction. Since the two side reactions are specified

on aniline, raising the reflux in COL-2 to sharpen this cut recovers that aniline into the product and holds the selectivity at the design value.

COL-1 is specified with a total condenser while its feed carries hydrogen, nitrogen and ammonia. A partial condenser with a vapour draw is the appropriate configuration for a feed containing non-condensables, and this is the first change to be made in Report 3.

ANRECOV returns 101.7 kmol/h of water to COL-1 with the recovered aniline, which is why the COL-1 distillate rate is set at 150 kmol/h and the reboiler duty at 11.08 MW. Routing this stream to DEC instead allows the decanter to remove the same water by phase separation at no energy cost.

Total heating in the flowsheet is 22.48 MW and total cooling 45.16 MW against a reaction exotherm of 17.57 MW, while the FEHE presently recovers 0.251 MW. The heat exchanger network design of Report 5 shall address this.

6 Objectives Which Could Not Be Accomplished, With Reasons

- The aniline–nitrobenzene binary pair has not yet been regressed, the laboratory time in this period having gone to building and converging the flowsheet. The pair bears directly on COL-2, whose reboiler runs at 144 °C while the databank parameters are valid over 83–99 °C, and the same shall be completed before the column design of Report 3.

7 Any Other Challenges

- MIX-1 has been left at a pressure specification of zero, so its outlet takes the lowest inlet pressure, which is the 0.3 bar of the COL-2 bottoms. A pump belongs at that point and shall be added with the equipment list.
- COL-1, with 20 stages and three-phase convergence, is the slowest block in the flowsheet and is re-solved on every recycle iteration. The same shall need attention when the loop is run repeatedly for the sensitivity studies of Report 4.

8 Self-Assessment

1. The reactor is an RStoic block at conversions specified by us, so the selectivity split and the 17.57 MW reaction duty are inputs rather than results. *Kinetic model in Report 4.*
2. The columns are converged but not yet designed. Stage counts, feed locations and reflux ratios have been chosen to obtain a working flowsheet rather than to meet the product specification. *Distillation column design and sequencing in Report 3.*
3. Utility duties are reported block by block, but no utility balance or cost has been drawn up and the heat integration is limited to the single feed–effluent exchanger. *Pinch analysis and utility costing in Report 5.*

9 Number of Hours Spent During This Period

Approximately **4 hours per member**, that is about **24 person-hours** for the team in this reporting period. This covers the property and operating-condition analysis, the process synthesis and the flow diagram, the Aspen Plus laboratory sessions in which the flowsheet was built and converged, the extraction of the balances and the preparation of this report.

10 Contributions From Individual Members

- **Nithin T M — flowsheet development in Aspen Plus.** Construction of the entire flowsheet block by block, the property set-up with the Henry component list and the regressed parameters, the reaction specification in RStoic, the configuration of the decanter and all three RadFrac columns, the closing of the three recycle loops, the purge specification and the convergence of the complete case.

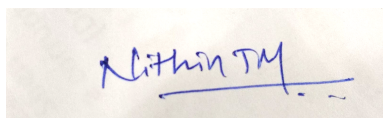
- **Sajag Masane — process synthesis and the process flow diagram.** The hierarchy of synthesis decisions, the recycle structure, the argument for taking the phase splits before distillation, the ordering of the separation train against the boiling point data, the placement of the feed–effluent exchanger, and the preparation of the process flow diagram.
- **Aditya Nitin Patil — material and energy balance.** Extraction of the converged stream and block results, the overall and component balances, the hydrogen loop and purge calculation, the energy balance with the reactor duty decomposition, and the closure checks against Report 1.
- **Nithin D H — component slate and operating conditions.** Compilation of the pure component boiling point data and the tabulation of the operating conditions with their bases.
- **Shubham Singh — column specifications.** Tabulation of the converged column specifications and results, and cross-checking of the reported condenser and reboiler duties.
- **Shivam Dongare — documentation.** Assembly of the figures from the Aspen Plus flowsheet, the reference list and the compilation of this report.

All the six members have reviewed the complete report and each of us is able to explain the sections owned by the others.

11 References

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8. Aspen Plus simulation file `Simulation1.che453.bkp`, with results exported as the stream results and model results summary workbooks, 21 August 2026.

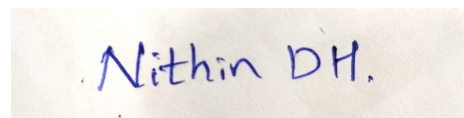
Signatures of members



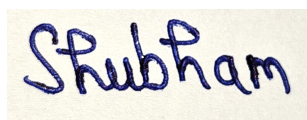
Nithin T M



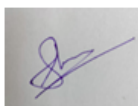
Aditya Nitin Patil



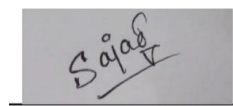
Nithin D H



Shubham Singh



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